Optimizing Chromium Low-Power Workloads on Intel Notebooks

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LPC 2022:

"Linux needs a Scheduler QOS API -- and it isn't nice(2)" Video PDF

Lively discussion in the session, and more in the hallways.

Vincent: *"Demonstrate value"*

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Acknowledgement

- Jianlin "Johnny" Qiu, Zheda Chen
- Srinivas Pandruvada, Rui Zhang, Noor Mubeen
- Tim C. Chen, Yu C. Chen, Rafael Wysocki

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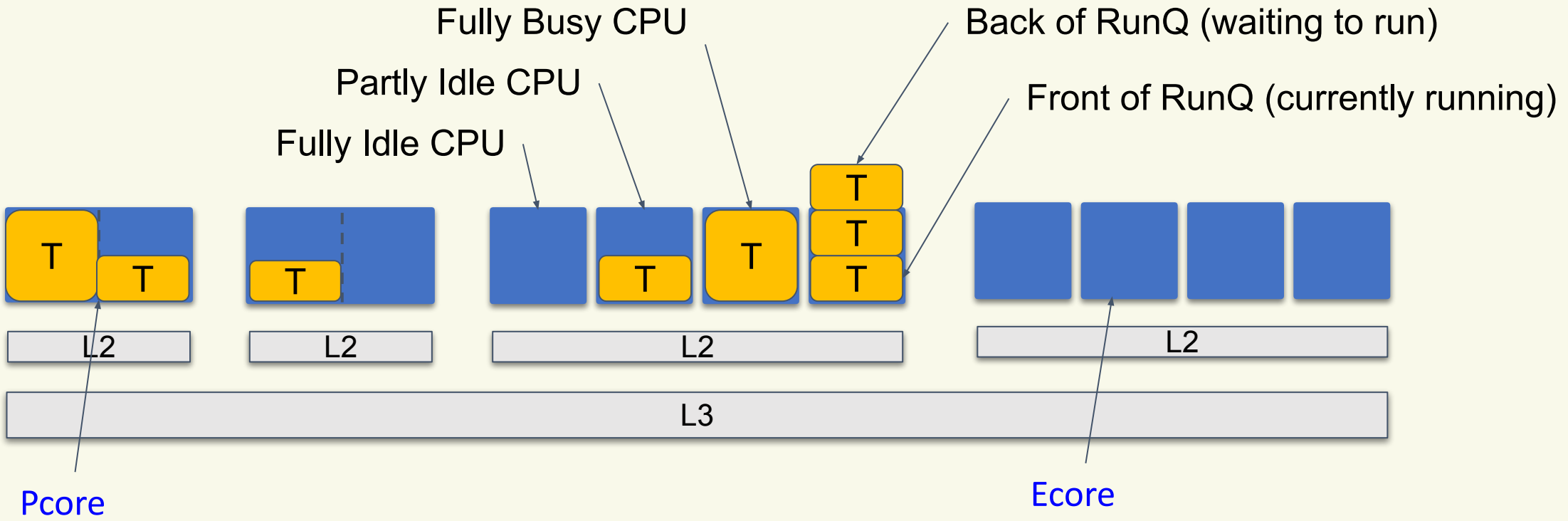
Linux Tasks on Intel Hybrid Topology



Uniform Instruction Set on all CPUs

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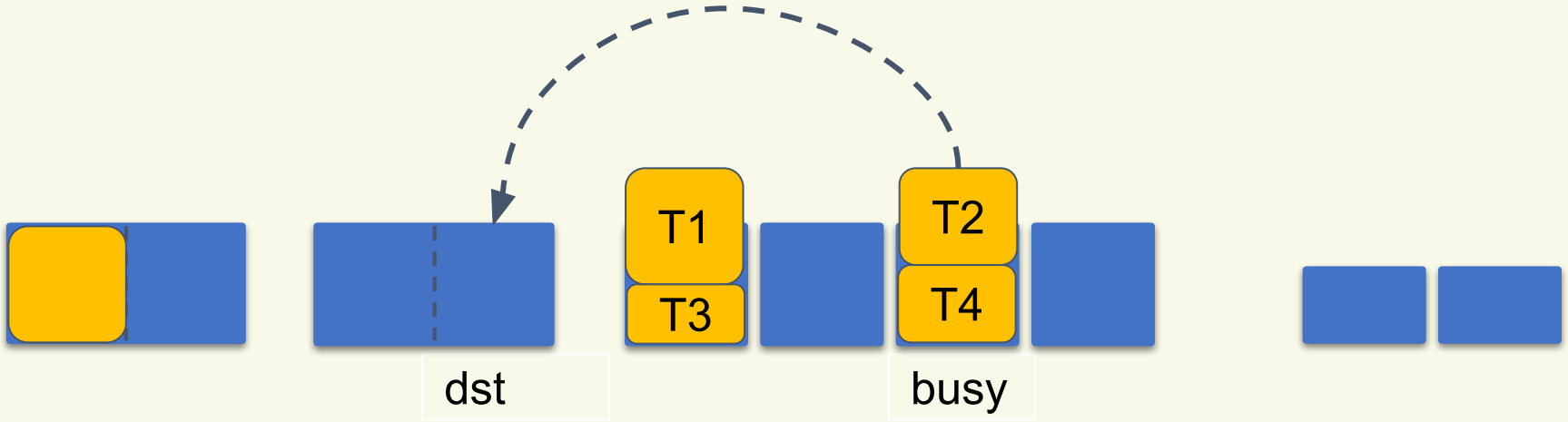
ITMT: Idle Load Balance (Ecore -> Pcore)

When: some CPUs fully idle (#task <= #CPU)

What: CPU "dst" enters idle, searches for "busy" to offload



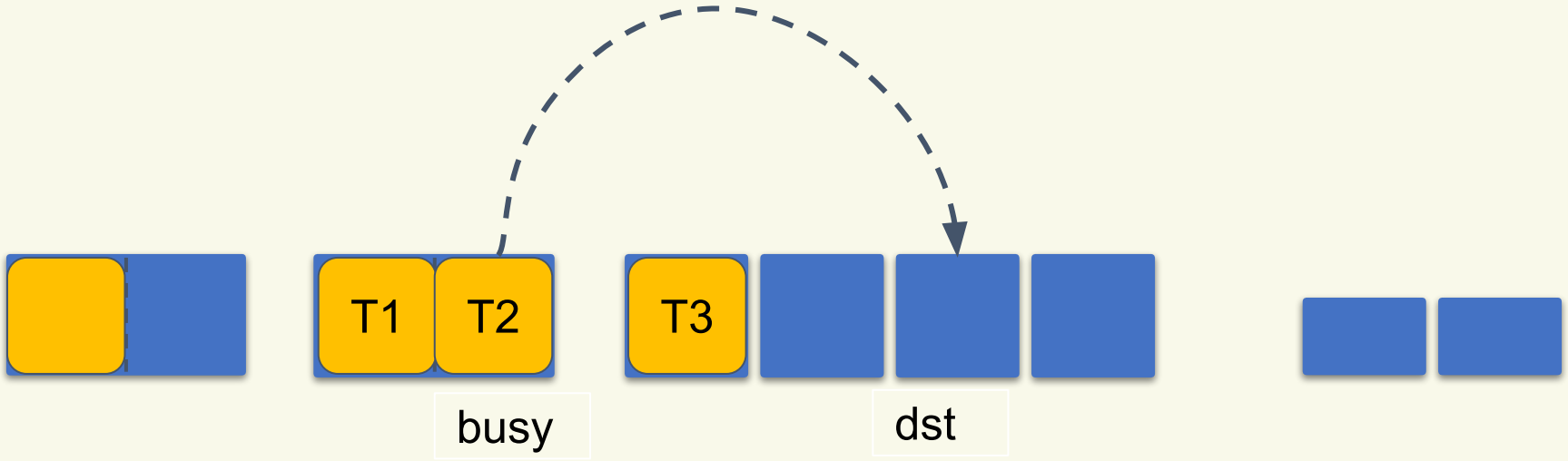
ITMT: fully-idle Pcore pulls from Ecore

* Other names and brands may be claimed as property of others

ITMT: Idle Load Balance (HT -> Ecore)

When: some CPUs fully idle (#task <= #CPU)

What: CPU "dst" enters idle, searches for "busy" to offload



ITMT: Ecore pull from busy-HT, always

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Energy Performance Preference (EPP)

Hardware P-States (HWP) is enabled by intel_pstate driver.

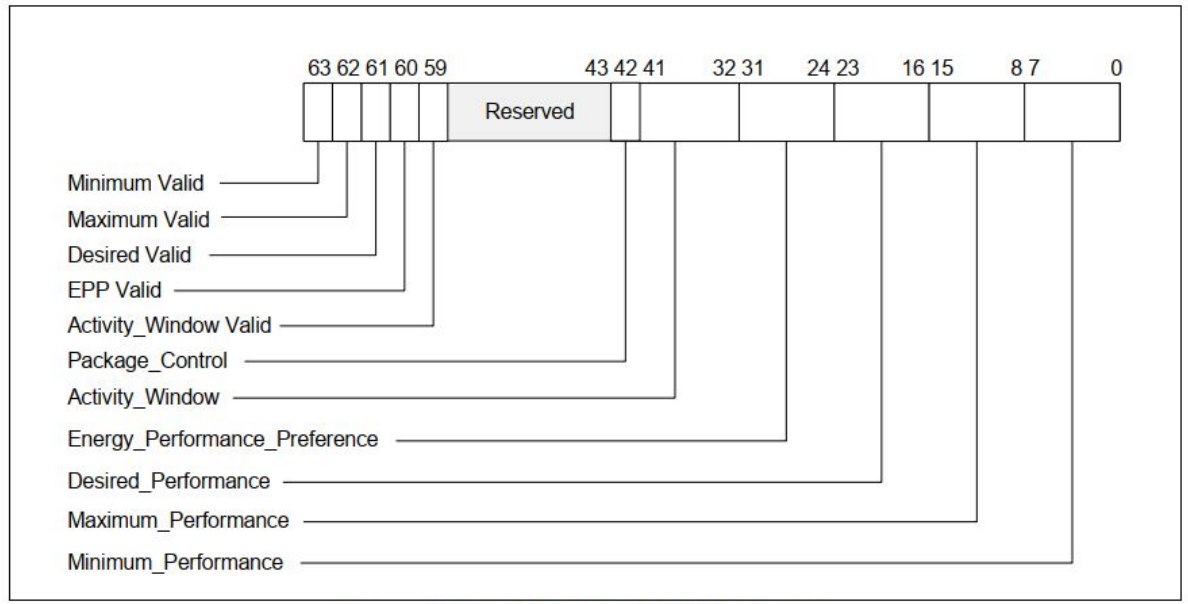

Figure 15-6. IA32_HWP_REQUEST Register

Per-CPU MSR.HWP.REQ.EPP is architectural HW interface but...

EPP numeric Value is Model Specific

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Linux intel_pstate driver maps Policy to EPP

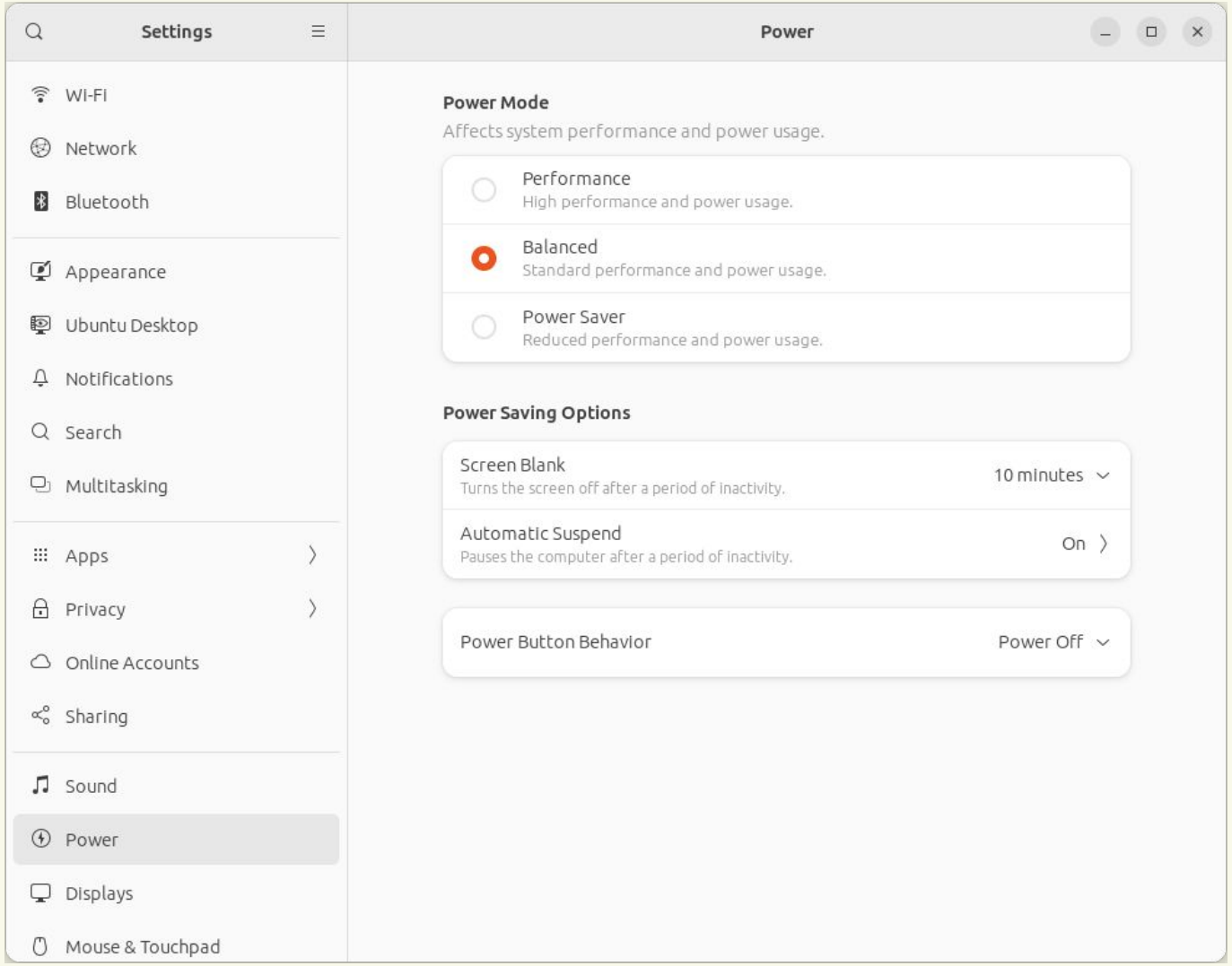
```
/sys/devices/system/cpu/cpu0/cpufreq$ grep . energy*
energy_performance_available_preferences:
    default
    performance
    balance_performance
    balance_power
    power
energy_performance_preference:
    balance_performance
```


* Other names and brands may be claimed as property of others

Linux Power Settings

Power Mode
  sets EPP
    on all CPUs

(to the same static value)



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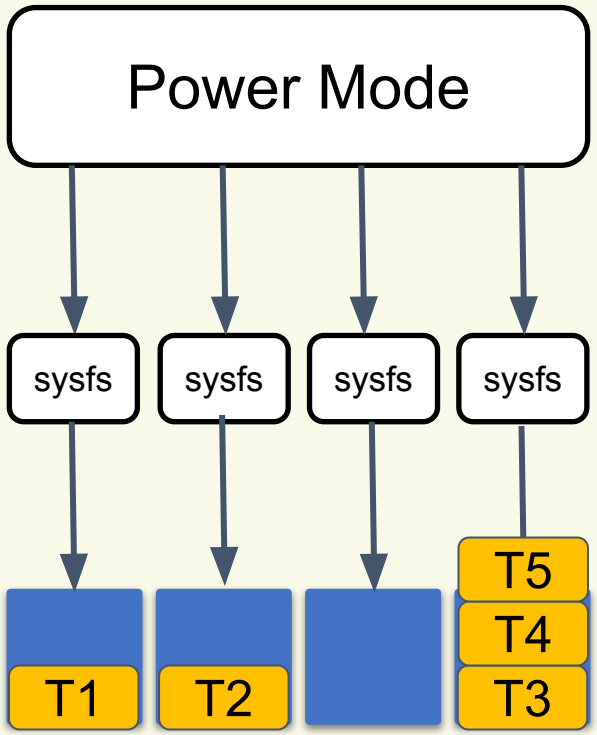
Per-CPU EPP - Limited Utility

1. If all tasks have same need

2. If diverse task are bound



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Chrome Browser Video Streaming Example

YouTube streaming 1080p 24fps video on chrome browser

System: Intel Core i7-12700H 6+8 (45W)

1x Ecore can retire this workload w/o performance impact

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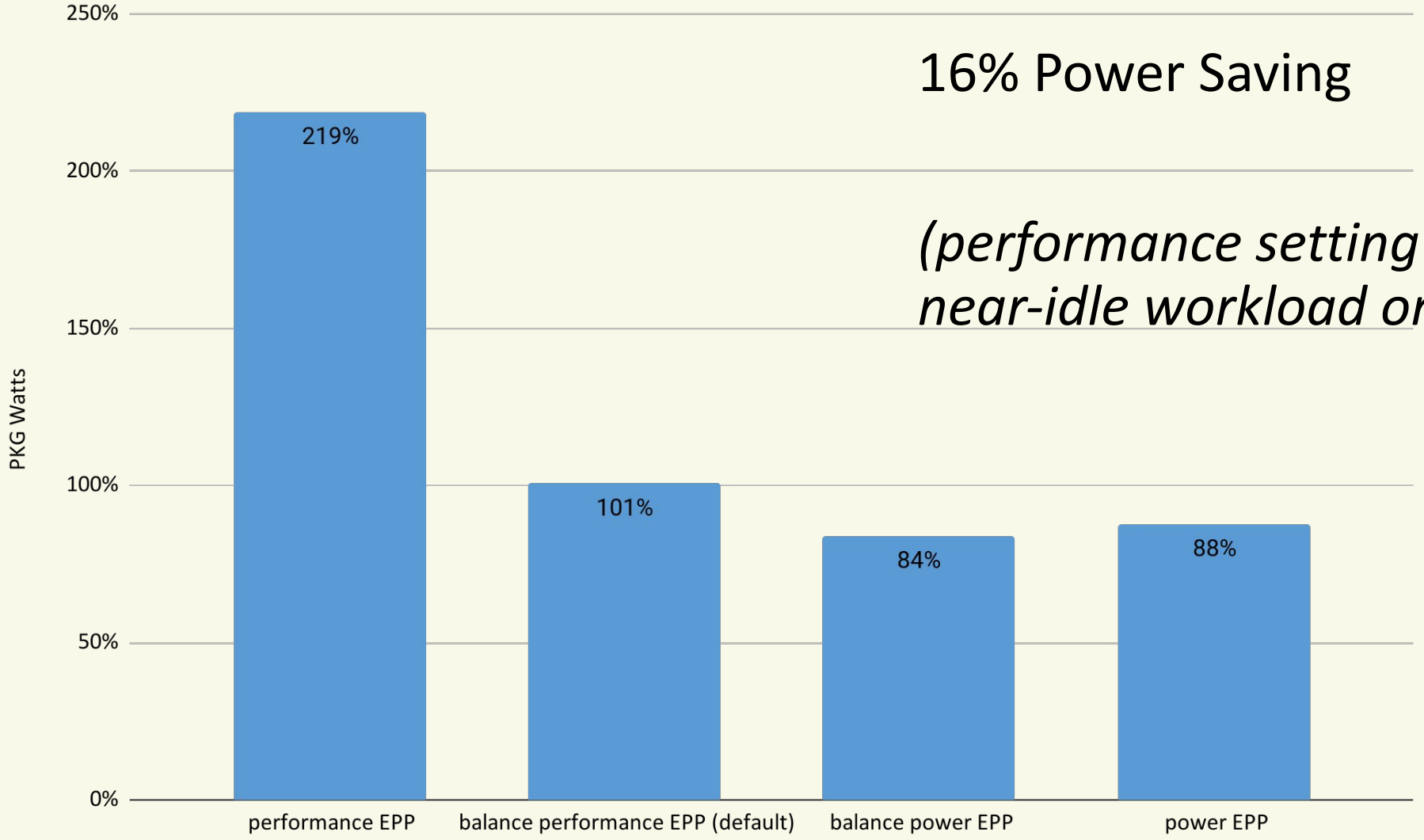
Video Streaming - All-CPU EPP Impact



16% Power Saving

(performance setting is toxic to near-idle workload on this CPU)

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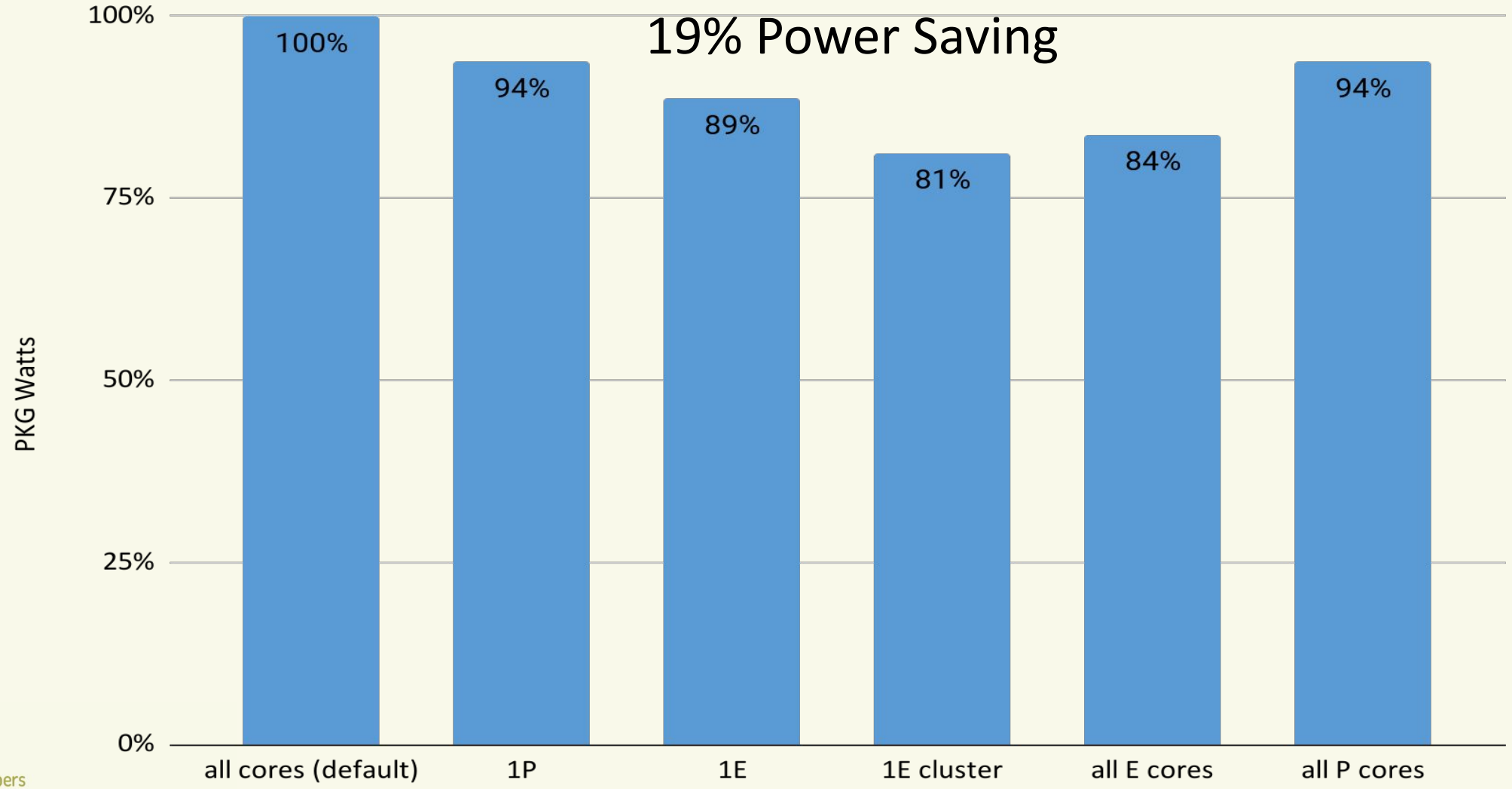
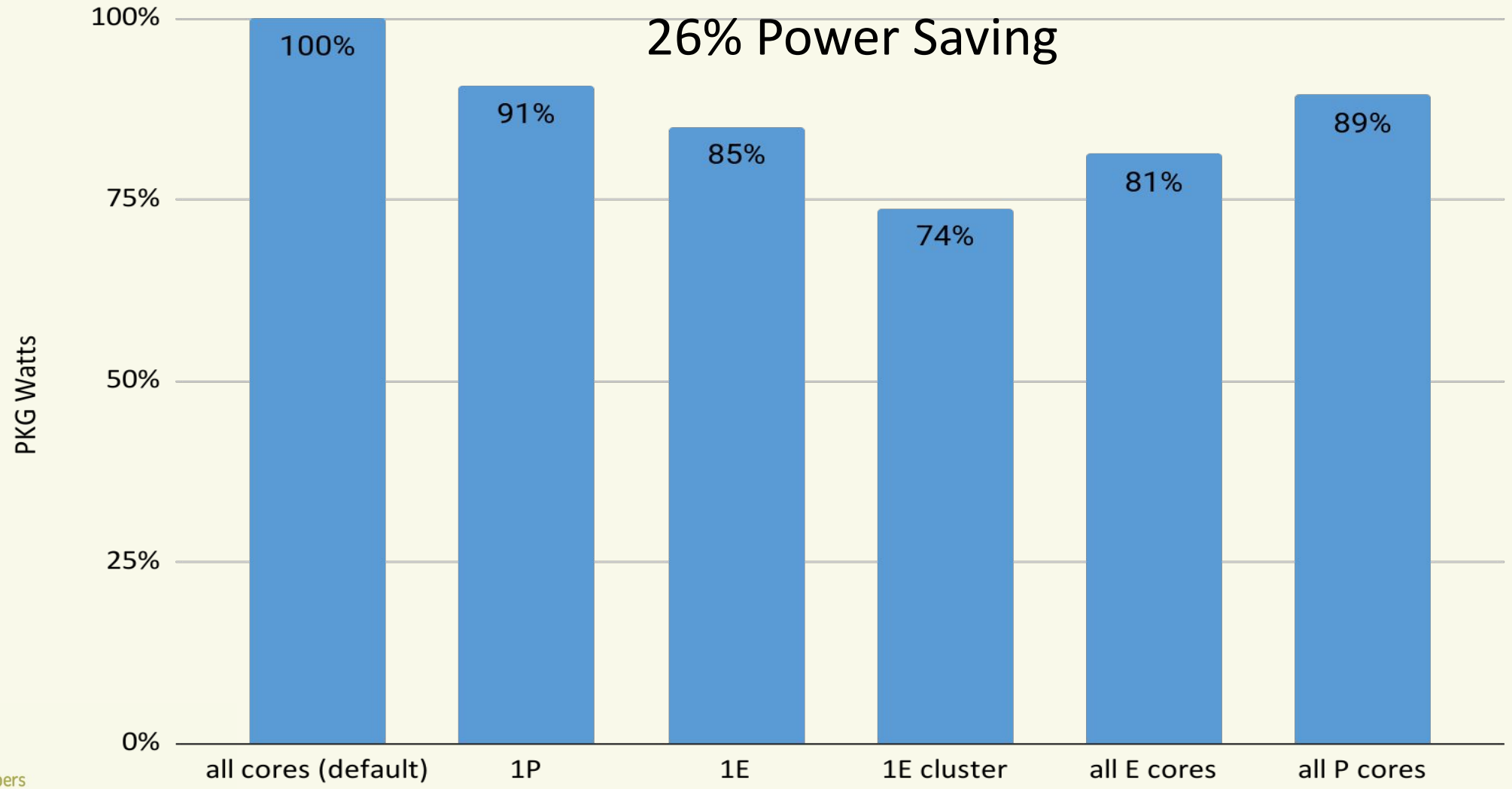
Video Streaming - Task Binding Impact



* Other names and brands may be claimed as property of others

Video Streaming - EPP + Task Binding Impact



* Other names and brands may be claimed as property of others

Chromium Video Conferencing Example

Google Video conference 2 party call

1080p 30fps (SW enc/dec)

System: Intel Core i7-12700H 6+8 (45W)

> 1x Ecore required to retire workload w/o performance impact

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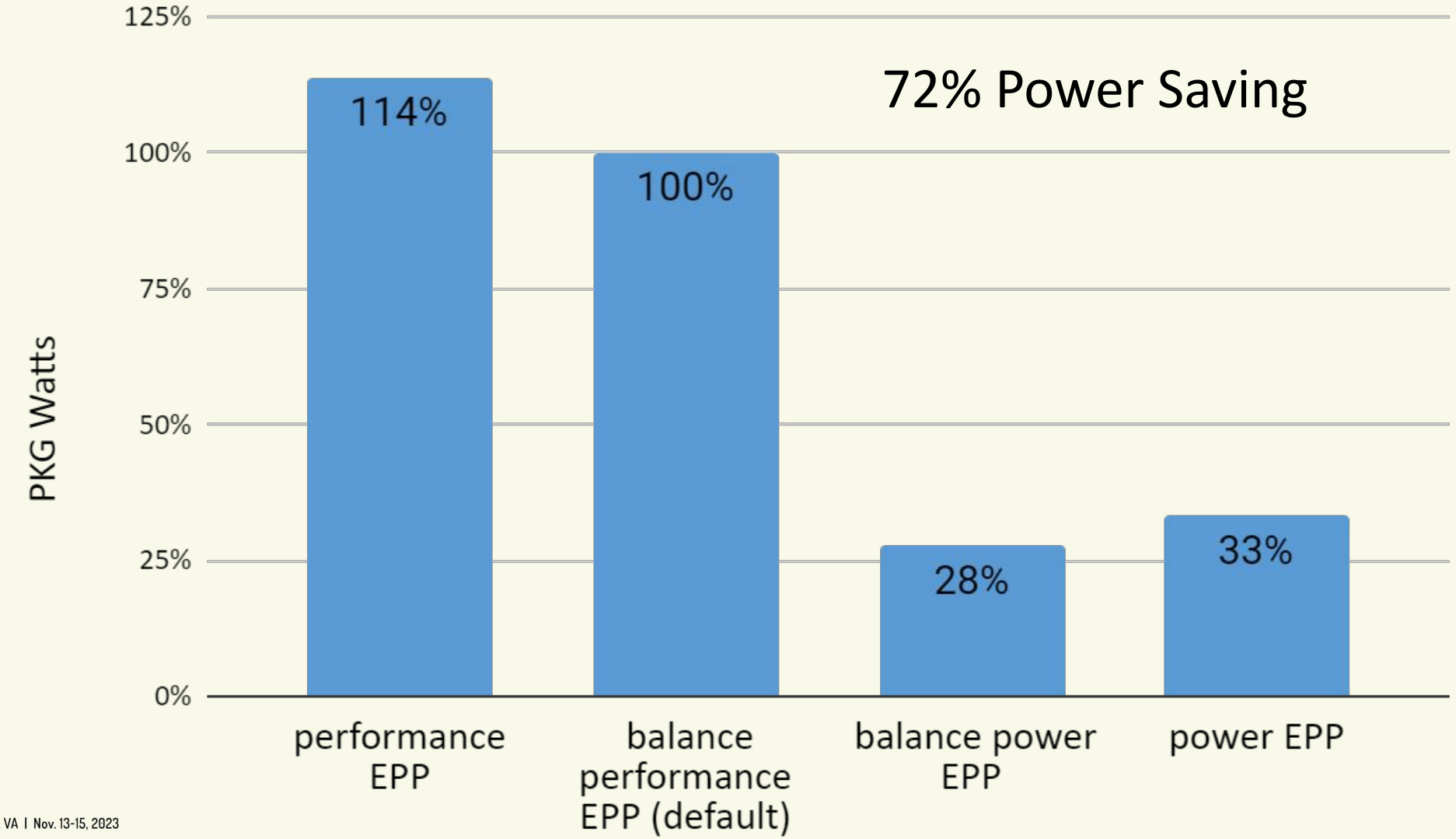
Video Conferencing - All-CPU EPP Impact



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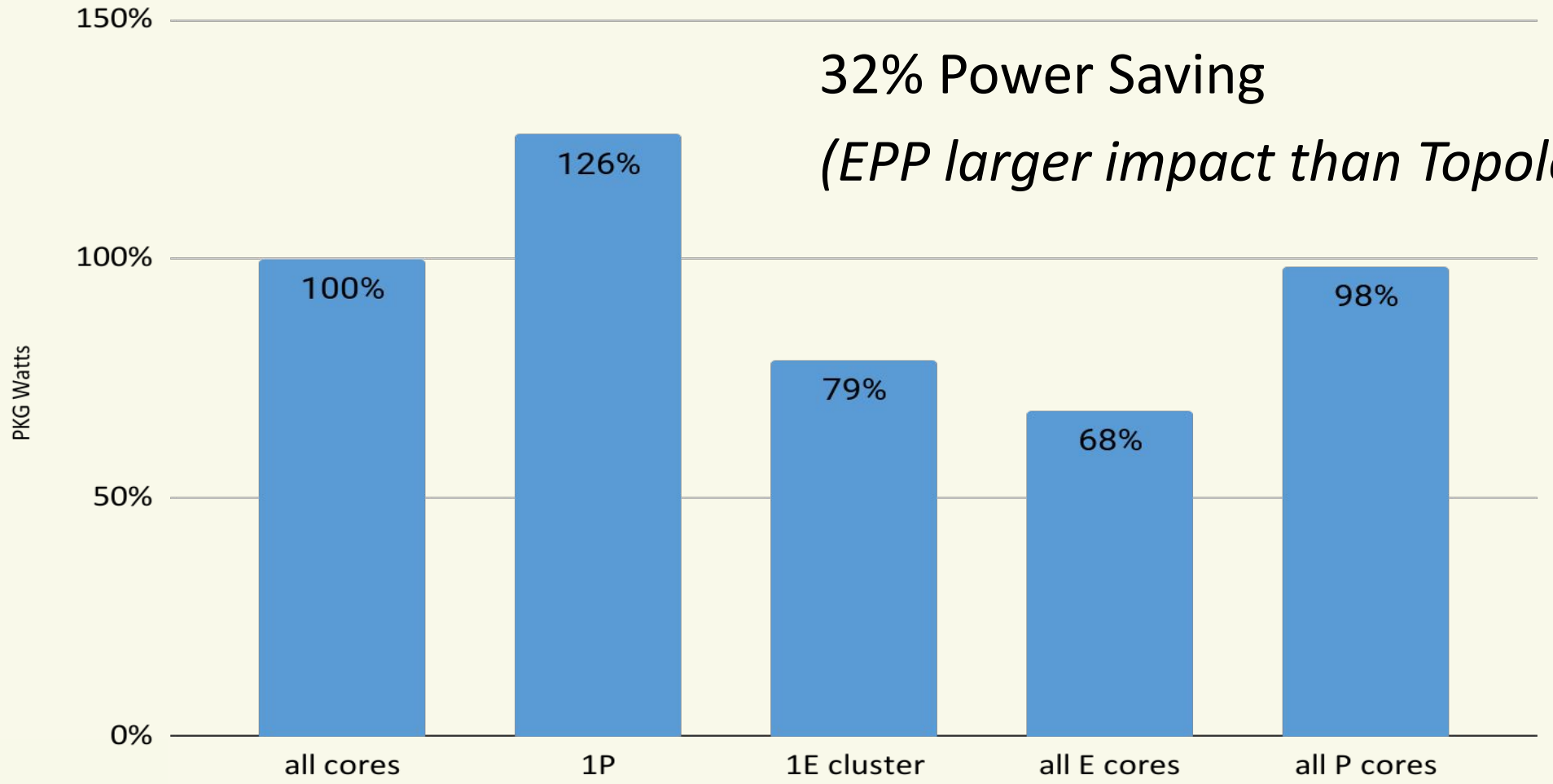
Video Conferencing - Task Binding Impact



32% Power Saving

(EPP larger impact than Topology)

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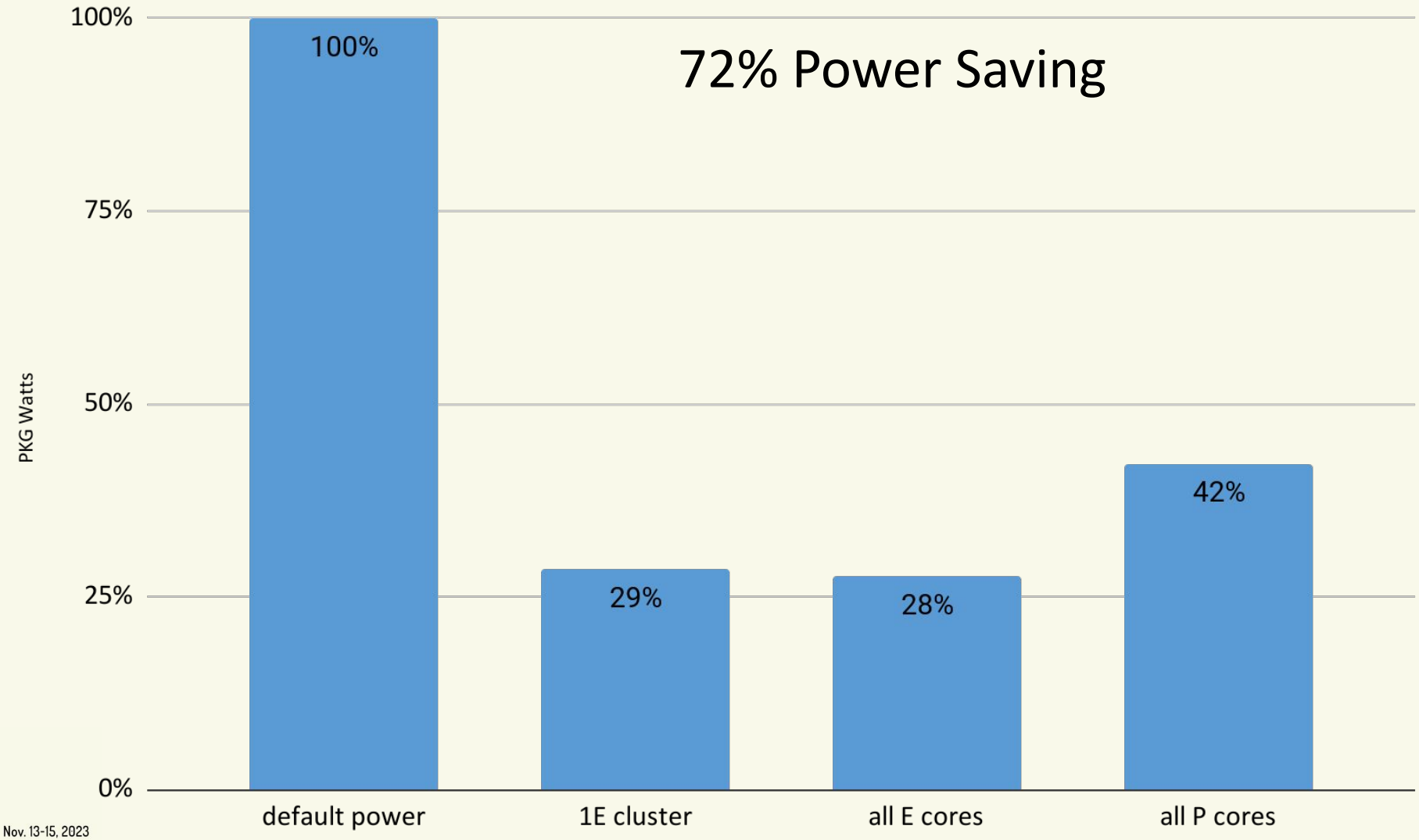
Video Conferencing - All-CPU EPP + Binding



* Other names and brands may be claimed as property of others

Is system wide orchestration the answer?

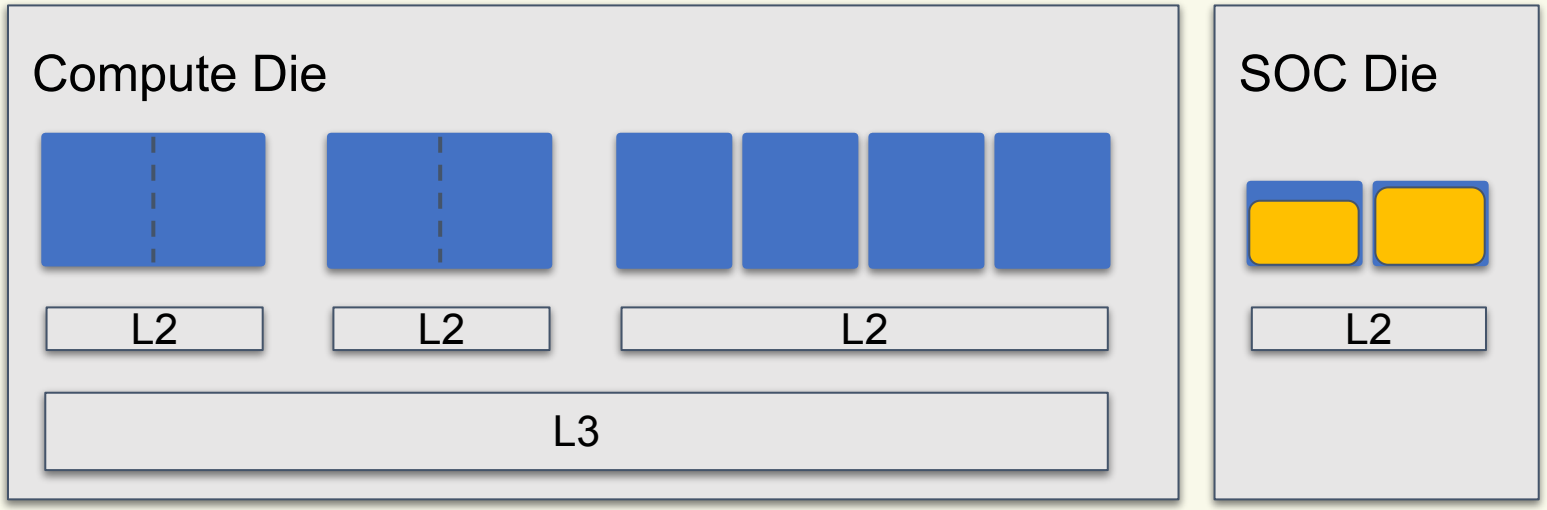
Sometimes yes.

- For near system-wide idle.
- When no risk to performance experience, can choose powersave

* Other names and brands may be claimed as property of others

Low Power Mode (eg. Meteor Lake)

1. No tasks should run on Compute Die CPUs
2. No IRQS should wake Compute Die CPUs
3. How much work can "fit" on the LP CPUs?



LP Mode: Compute die is 100% idle, all work runs on SOC Die

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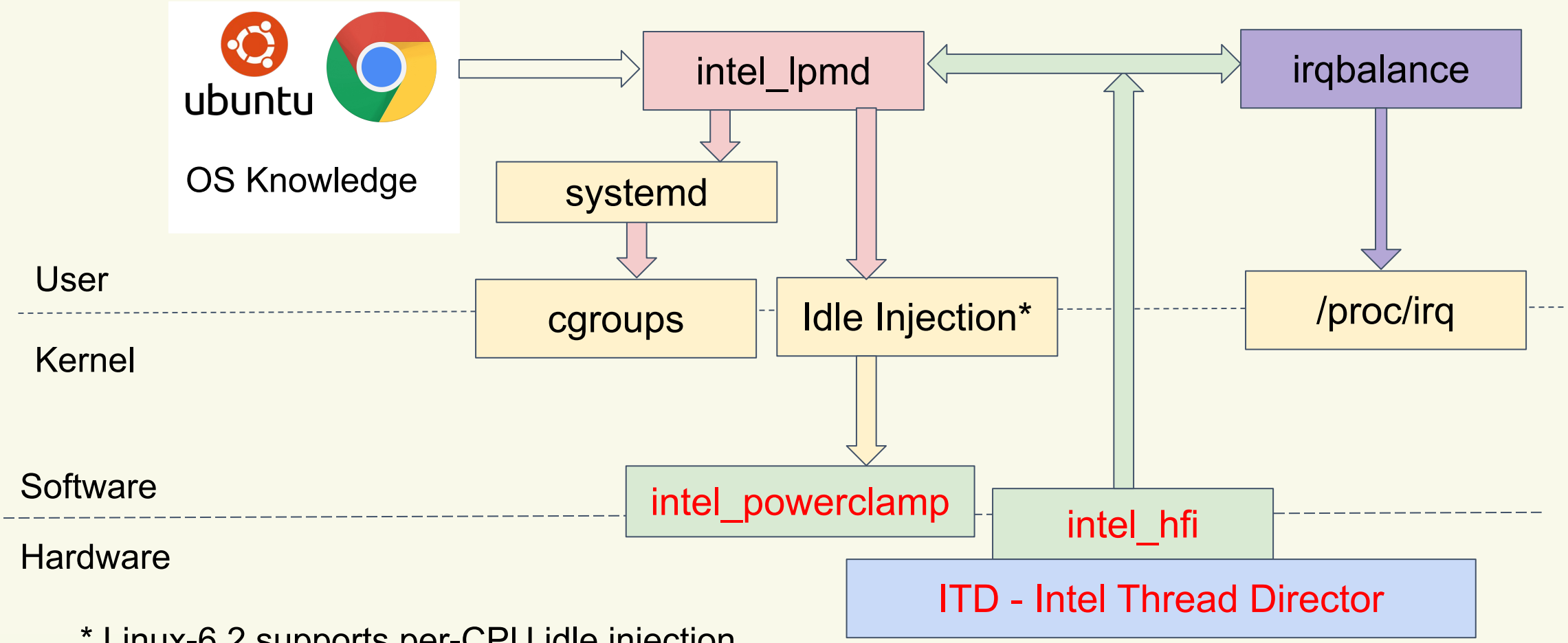
Low Power Mode Policy in User-Space



* Linux-6.2 supports per-CPU idle injection

* Other names and brands may be claimed as property of others

Is system wide orchestration the answer?

Sometimes no.

- When diversity of task needs, especially within a process.
- When load high enough that power-bias degrades performance.

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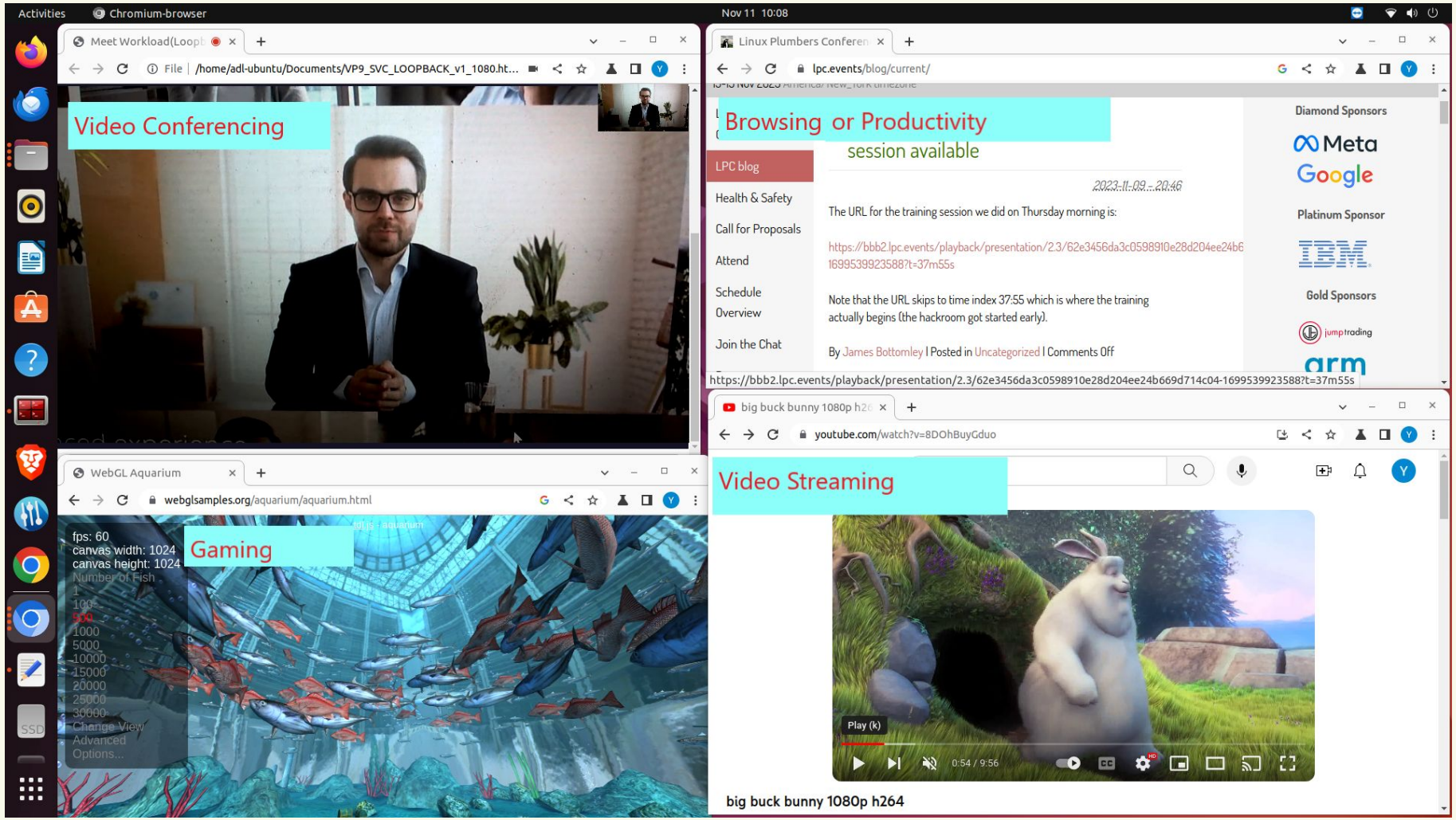
Simultaneous Diverse Task requirements



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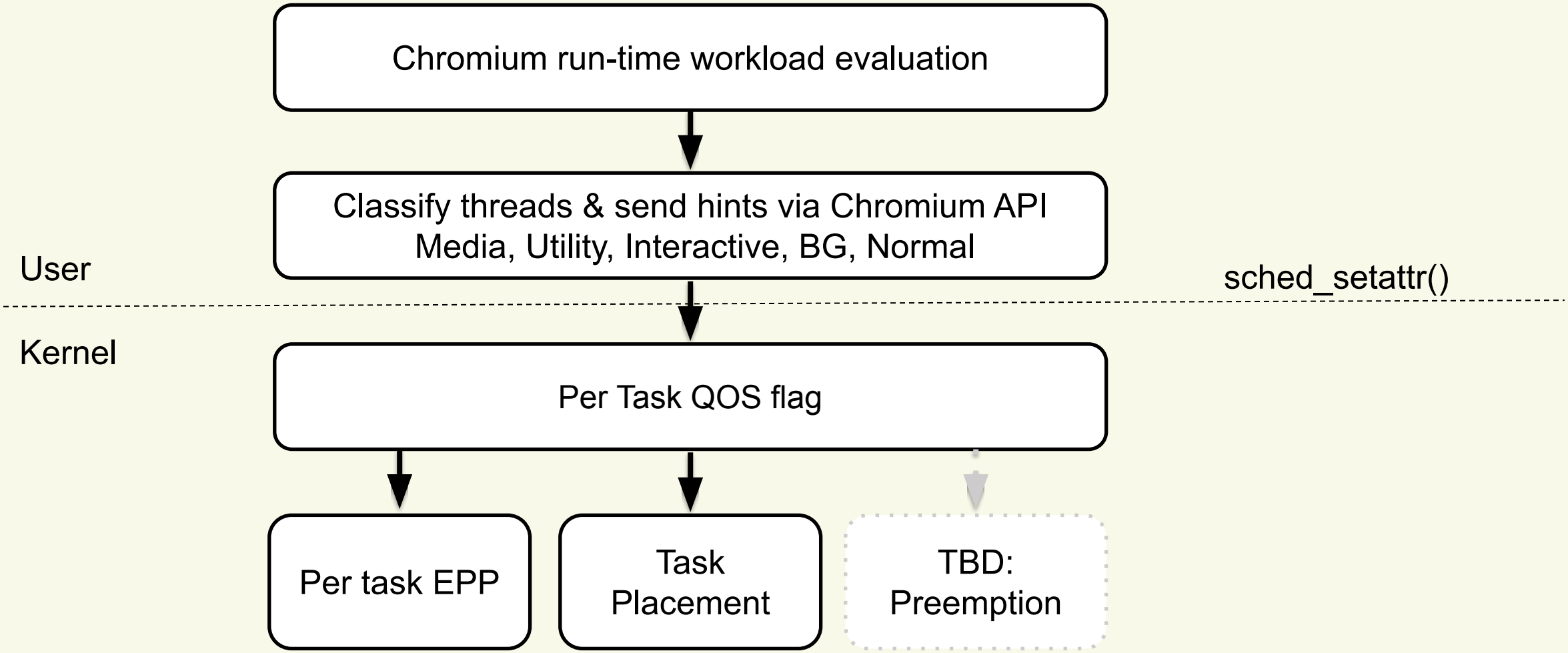
Browser Requests per-task Power vs. Perf Policy



* Other names and brands may be claimed as property of others

EQOS via per-task EPP

New QOS member in task_struct

Change EPP of CPU upon sched_set_attr() and context switch

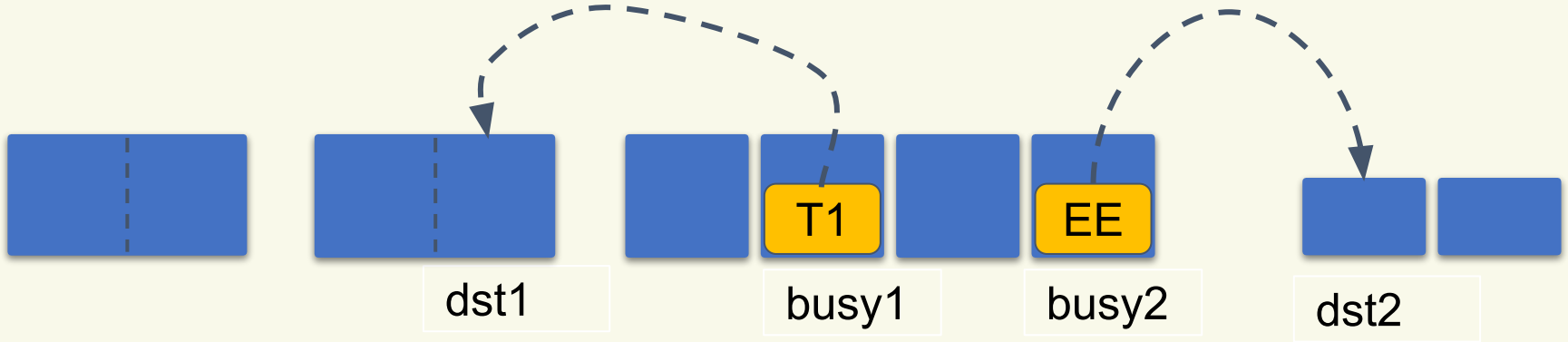
Yield to use of legacy sysfs interface

* Other names and brands may be claimed as property of others

EQOS via Task Placement

When: partially idle

What: Default tasks prefer high priority, EE tasks prefer low priority



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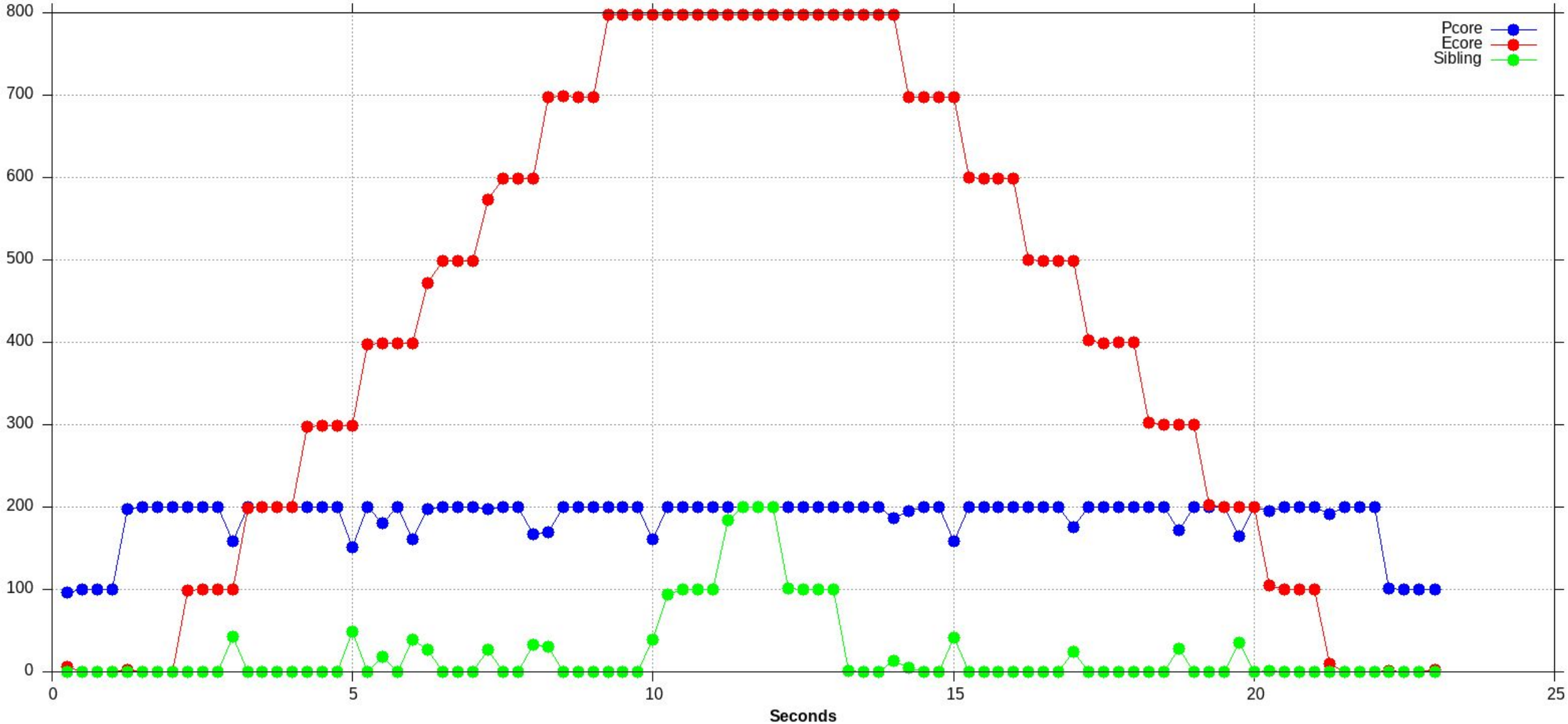
Utilization by CPU-Type

GETCPU@100

ADL-P_4 4xP@4.7 8xE@3.5 9W

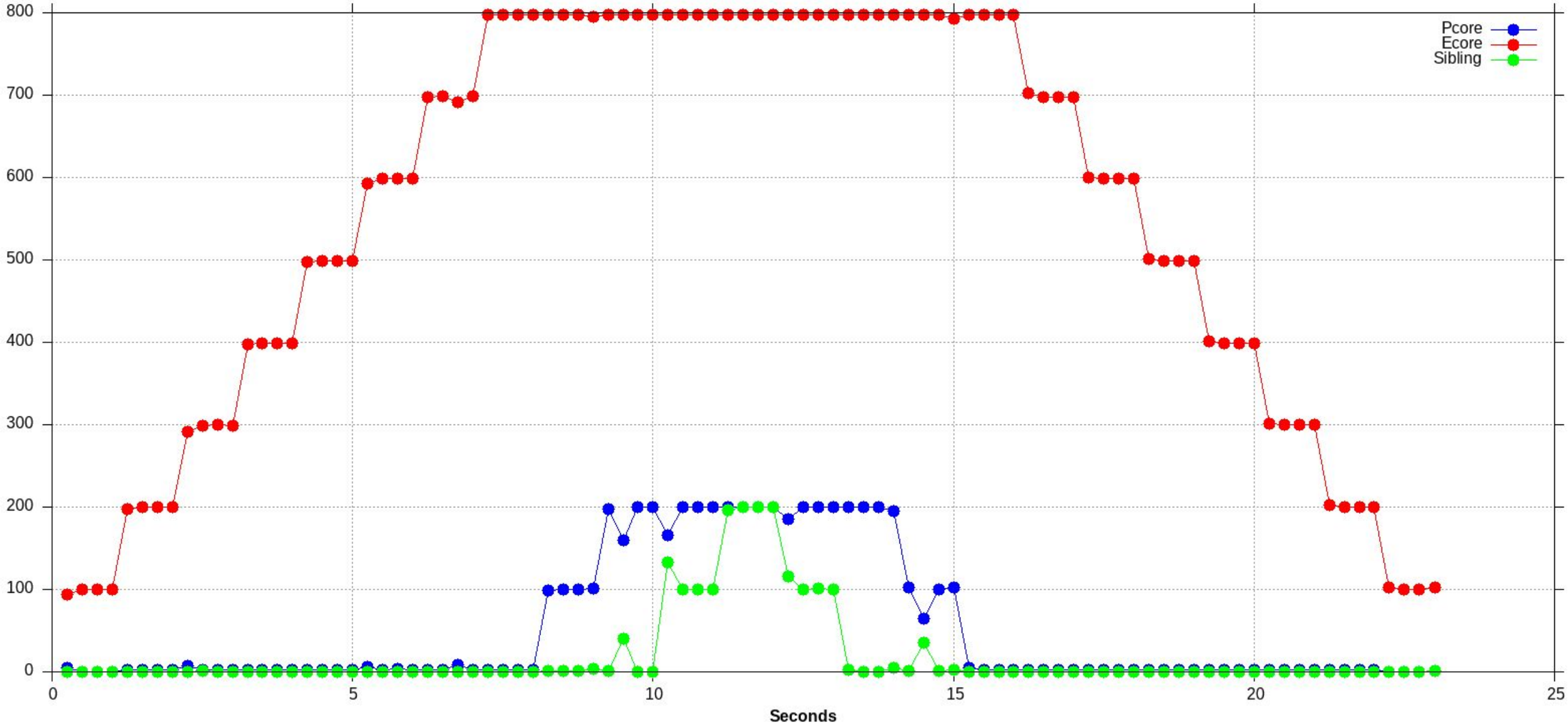


pyramid100-GETCPUx12-20231109_203936
yogini 2023.10.23 - Len Brown <len.brown@intel.com>

Utilization by CPU-Type

GETCPU-ME@100

ADL-P_4 4xP@4.7 8xE@3.5 9W



pyramid,EQOS_MAX_EFFICIENCY100-GETCPUx12-20231109_203936

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Chromium Demo using EQOS API

Statically assigned just Chromium's Render task to:

EQOS_BALANCE_EFFICIENCY

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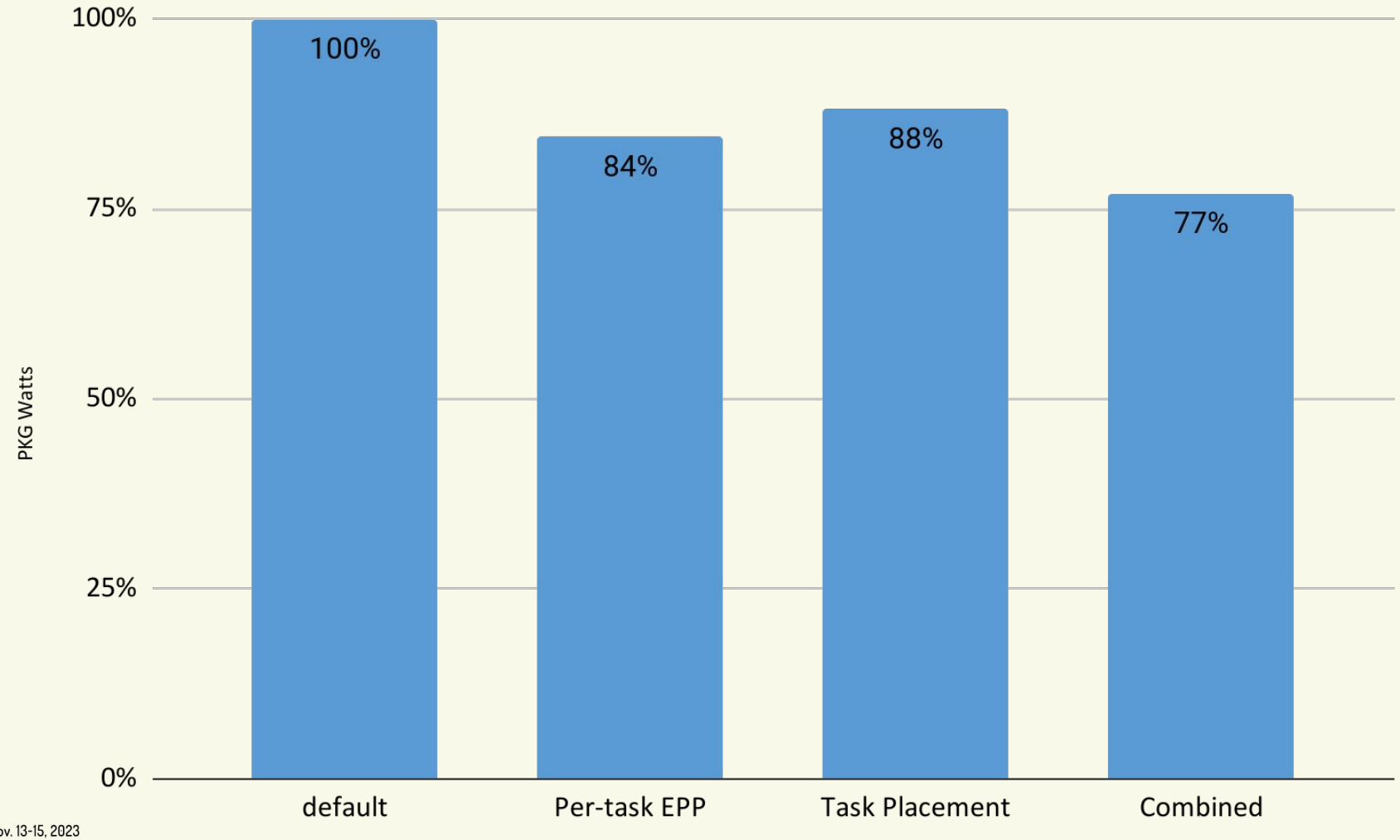
Chromium Video Streaming using EQOS API

23% Power Saving



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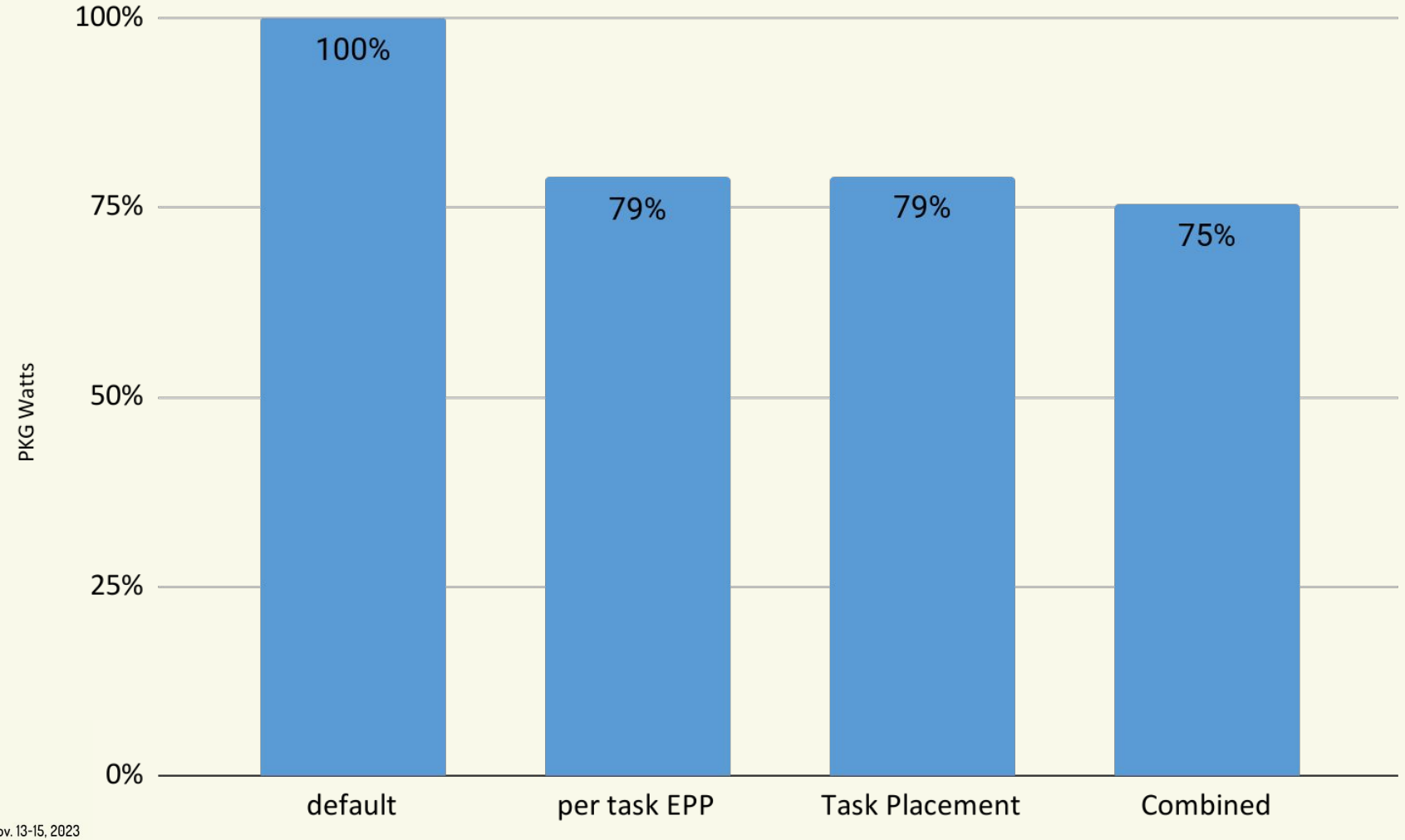
Chromium Video Conf with EQOS API

25% Power Saving



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Yet to Do

Chromium statically assigns QOS per-task type

- Better done dynamically based on user experience
- (eg. don't opt into EE if you are dropping frames)

Preemption can benefit both Perf and EE tasks.

- Orthogonal to EQOS, likely complementary

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Limitation: EQOS_LB depends on ILB

if no idle, then no ILB

periodic LB

Short running tasks missed by ILB

Tasks could wake up on preferred CPU from the start

hook into the wakeup path

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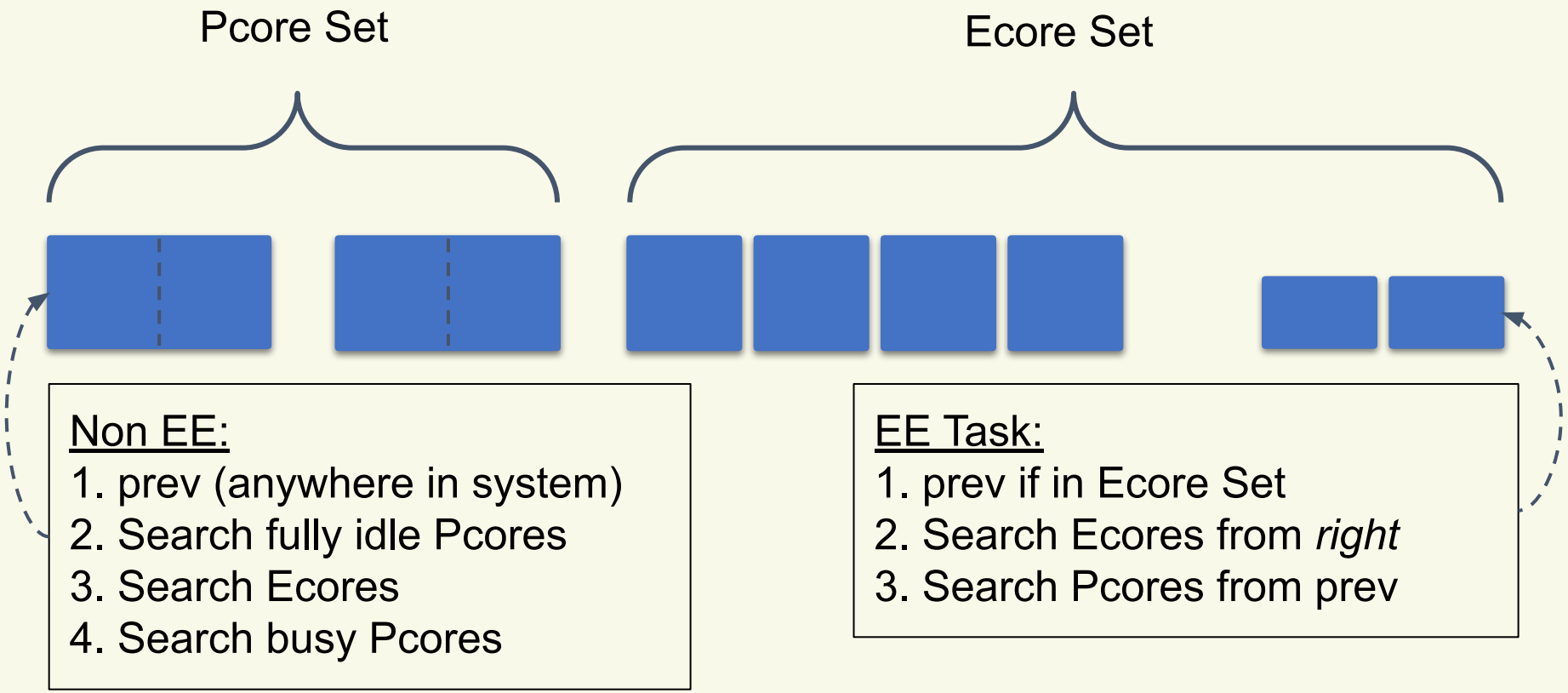
EQOS: Wake-up Idle CPU Search [RFC]

- Pcore Set (includes ITBM 3.0 "favored" cores)
- Ecore Set (includes Lcores)



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Code is Here

https://github.com/ricardon/linux/tree/rneri/eqos-demo

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